NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

SEMESTER: I

DEPARTMENT

OF

COMPUTER SCIENCE AND ENGINEERING

PRACTICE WORK BOOK

SUBJECT CODE: BE01R00021

SUBJECT NAME: PHYSICS

Subject Code:	BE01R00021	Subject Name	Credit:	4
Semester:	I	PHYSICS	Hours:	120
Department:		Computer Science and Engineering		

CHAPTER: 1

PROPERTIES OF MATTER

- Stress – Strain – Hooke's law –
- Elastic Behaviour of Material –
- Young's modulus by cantilever depression –
- Non-uniform bending –
- Uniform bending-
- Application -I-shaped girders.
- Torsional Pendulum – Couple per unit twist of a wire, Timeperiod, Application- Determination of Rigidity Modulus.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Explain the term bulk modulus, modulus of rigidity and yield point. **(Jan-2024)[NLJIET]**	3	CO1	UN
2.	Discuss stress-strain diagram. **(Mar-2023)[NLJIET]**	3	CO1	UN
3.	Define Hooke's law, stress and strain. **(Mar-2022)[NLJIET]**	3	CO1	RM
4.	Define: (1) Stress (2) Strain and (3) Young's modulus. **(Jan-2025-New) [NLJIET]**	3	CO1	RM
5.	Derive the formula for time period of a torsional pendulum. **(Jan-2024)[NLJIET]**	4	CO1	AN
6.	Define and describe Elasticity and Plasticity in detail by giving suitable examples. **(Mar-2021)[NLJIET]**	4	CO1	RM
7.	Draw Stress versus Strain diagram with necessary notation and discuss it. **(Jul-2024)[NLJIET]**	4	CO1	UN
8.	Draw stress- strain diagram. Explain the main points of it. **(Jan-2020)[NLJIET]**	4	CO1	UN
9.	Derive the equation that relates young's modulus to the bending of a beam in case of non-uniform bending. **[NLJIET]**	4	CO1	AN

10.	Derive the equation that relates young's modulus to the bending of a beam in case of uniform bending. **[NLJIET]**	**4**	**CO1**	**AN**
11.	Write a short note in I-shaped girder. **[NLJIET]**	**4**	**CO1**	**AP**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	What is torsional pendulum? Derive expression of time period for vibration of torsional pendulum. **(Jul-2024)[NLJIET]**	7	**CO1**	**AN**
2.	Derive the expression for depression of Cantilever loaded at free end. **(Jul-2024)[NLJIET]**	7	**CO1**	**AN**
3.	What is Cantilever? Obtain the expression for depression at free end of thin beam clamped horizontally at one end and loaded at other end. **(Jan-2024)[NLJIET]**	7	**CO1**	**AN**
4.	Derive an expression for the depression of the free end of cantilever. What are the applications of cantilevers? **(Mar-2023)[NLJIET]**	7	**CO1**	**AN**
5.	Draw stress versus strain graph with appropriate notations. Explain elastic limit and upper yield point in detail. **(Aug-2022)[NLJIET]**	7	**CO1**	**UN**
6.	Describe Stress-Strain diagram in detail. **(Mar-2022)[NLJIET]**	7	**CO1**	**AN**
7.	Discuss in detail the different types of elasticity. List different factors affecting elasticity.**(Mar-2022)[NLJIET]**	7	**CO1**	**UN**
8.	Explain the terms: (i) Young's modulus, (ii) Bulk modulus, (iii) Shear modulus and (iv) poison's ratio. Show any one relation between different moduli of elasticity. **(Mar-2021)[NLJIET]**	7	**CO1**	**UN**
9.	Draw: Stress – Strain diagram with necessary notation. Explain: Elastic Limit and Upper Yield Point in detail. **(Mar-2021) [NLJIET]**	7	**CO1**	**UN**

10.	What is Cantilever? Obtain the expression for depression at free end of thin beam clamped horizontally at one end and loaded at other end. **(Mar-2021) (Jan-2025-New) [NLJIET]**	**7**	**CO1**	**AN**
11.	Derive an expression for depression of cantilever. **(Aug-2023) [NLJIET]**	**7**	**CO1**	**AN**
12.	Explain Young's Modulus, shear modulus, bulk modulus and Poisson's ratio. **(Jan-2020) [NLJIET]**	**7**	**CO1**	**UN**
13.	Explain in detail Stress – Strain diagram for a bar or wire. **(Jun-2019) [NLJIET]**	**7**	**CO1**	**UN**
14.	Explain torsional pendulum in brief with suitable diagram and derive the equation of twisting couple per unit twist of a wire. **(Jun-2025-New) [NLJIET]**	**7**	**CO1**	**AN**
15.	Explain: (i) Elastic hysteresis, (ii) Elastic after-effect, and (iii) Elastic fatigue. **(Jun-2025-New) [NLJIET]**	**7**	**CO1**	**UN**
	EXAMPLES			
1.	An elastic rod having diameter of 30 mm, 10 cm long extends by 2.5 cm under tensile load of 28 kN. Find the stress, strain and the Young's modulus for the material of the rod. **(Jun-2019) [NLJIET]**	**3**	**CO1**	**AP**
2.	A load of 2.6 kg produces an extension of 1 mm in a wire of 3.082 m and 1 mm in diameter. Calculate the Young's modulus of a wire. **(Mar-2021) [NLJIET]**	**3**	**CO1**	**AP**
3.	A steel wire has 3 m length and 0.5 mm radius. When it is stretched by force of 49 N find (i) longitudinal stress, (ii) longitudinal strain and (iii) elongation produced in the wire if Young's modulus of steel is 2.1 x 10^{11} N/m^2. Take g = 9.8 m/s^2. **(Aug-2022)[NLJIET]**	**3**	**CO1**	**AP**
4.	A brass bar having a cross-section of 1 cm^2 is supported on two knife-edges 1.5 m apart. A load of 2 kg at the center of the bar depresses that point by 2.75 mm. What is the Young's Modulus for brass? **(Mar-2022) [NLJIET]**	**3**	**CO1**	**AP**

5.	An elastic rod having diameter of 30 mm, 10 cm long extends by 2.5 cm under tensile load of 28 kN. Find the stress, strain and the Young's modulus for the material of the rod. **(Jan-2024) [NLJIET]**	**3**	**CO1**	**AP**
6.	A solid disc of 1 kg mass and 0.2 m diameter is suspended in a horizontal plane by a vertical wire attached to its center. The length and diameter of the wire is 1.5 m and 2 mm, respectively. Calculate modulus of rigidity of wire and the time period Torsional oscillations if Torsional rigidity CS = 7.8×10^{-3} kg·m^2 /s^2. **(Jun-2019) [NLJIET]**	**4**	**CO1**	**AP**
7.	(ii) What force is required to stretch a steel wire to double the length when its area of cross section is 1 cm^2. Given that Young's modulus of wire is 7×10^{10} N/m^2.**(Jan-2020) [NLJIET]**	**4**	**CO1**	**AP**
8.	(i) Calculate the twisting couple of a solid shaft of 2 m length and 0.2 m diameter when it is twisted through an angle of 0.01 rad. The coefficient of rigidity of the material is 8 x 10^{10} N/m^2. (ii) Calculate the twist produced in a solid wire of 5 m length and 0.1 m radius. The twisting couple produced in the wire is 2 x 10^4 N m and the coefficient of rigidity of the material is 10^{11} N/m^2 . **(Aug-2022) [NLJIET]**	**4**	**CO1**	**AP**
9.	A wire of length 2 m and radius 0.5 mm is clamped at one of its ends. Calculate the torque required to twist the other end by 90°. The rigidity modulus of the wire is 2.8 x 10^{10} N/m^2. **(Mar-2023) [NLJIET]**	**4**	**CO1**	**AP**
10.	Find the force required to increase the length of a wire of 10^{-6} m^2 area of cross-section by 50 % whose Young's modulus is 2×10^{11} N/m^2. **(Jun-2025-New) [NLJIET]**	**4**	**CO1**	**AP**

CHAPTER: 2

WAVES, MOTION AND ACOUSTICS

- Simple Harmonic motion, Free, forced, Resonance,
- Damped and undamped vibration,
- Damped harmonic motion,
- Force vibration and amplitude resonance,
- Velocity resonance and energy intake,
- Wave motion, transverse and longitudinal vibration,
- Sound absorption and reverberation,
- Sabine's formula and usage (excluding derivation),
- Acoustic of building,
- Ultrasonic waves - Properties - Generation – Piezoelectricmethod – Detection- Kundt's tube
- Application of Ultrasonics in industries – NDT.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	What is Ultrasound? List various methods of detecting ultrasonic Waves. **(Jun-2019) [NLJIET]**	2	CO2	RM
2.	Define wave motion. Discuss different types of waves. **(Mar-2022) [NLJIET]**	3	CO2	UN
3.	Define simple harmonic motion (SHM). Give Any two examples of it. **(Aug-2023) [NLJIET]**	3	CO2	RM
4.	Define: (i) Transverse waves, (ii) Ultrasonic waves, and (iii) Damped oscillations. **(Jun-2025-New) [NLJIET]**	3	CO2	RM
5.	Write short note on Reverberation and Reverberation time. **(Jun-2019) [NLJIET]**	3	CO2	UN
6.	Explain sound absorption and reverberation. **(Jan-2024) [NLJIET]**	3	CO2	UN

7.	Define resonance in an oscillating system. **(Jan-2020) [NLJIET]**	**3**	**CO2**	**RM**
8.	Classify sound on the basis of frequency with suitable examples. **(Jan-2020) [NLJIET]**	**3**	**CO2**	**UN**
9.	Give classification of sound waves based on frequency of wave. **(July-2024) [NLJIET]**	**3**	**CO2**	**UN**
10.	Describe any three properties of ultrasonic sound. **(Mar-2021) [NLJIET]**	**3**	**CO2**	**RM**
11.	Write the properties of ultrasound wave. **(Mar-2023) [NLJIET]**	**3**	**CO2**	**RM**
12.	Describe any three properties of ultrasonic sound. **(Jan-2024) [NLJIET]**	**3**	**CO2**	**RM**
13.	Write any three properties of ultrasonic waves. **(July-2024) [NLJIET]**	**3**	**CO2**	**RM**
14.	Define ultrasonic waves. Mention two applications of Ultrasonic waves. **(Aug-2022) [NLJIET]**	**3**	**CO2**	**AP**
15.	Define the term: (1) Reverberation time (2) Piezoelectric effect (3) Acoustic grating. **(Mar-2023) [NLJIET]**	**3**	**CO2**	**RM**
16.	Define:(1) Reverberation (2) Nano-materials (3) Absorption co-efficient **(Jan-2025-New) [NLJIET]**	**3**	**CO2**	**RM**
17.	Define: (i) Reverberation, (ii) Echo, and (iii) Echelon effect. **(Jun-2025-New) [NLJIET]**	**3**	**CO2**	**RM**
18.	What is reverberation and reverberation time? Write Sabin's formula for reverberation time and explain each term. **(July-2024) [NLJIET]**	**3**	**CO2**	**UN**
19.	Write down various applications of ultrasonic waves. **(Jan-2020) [NLJIET]**	**3**	**CO2**	**AP**
20.	Explain NDT. Discuss advantages of NDT. **(Jan-2024) [NLJIET]**	**3**	**CO2**	**UN**
21.	Explain NDT with it's objectives. **(Mar-2022) [NLJIET]**	**4**	**CO2**	**UN**

22.	Define and differentiate between transverse and longitudinal waves. **(Mar-2021) [NLJIET]**	**4**	**CO2**	**UN**
23.	Differentiate Free and Forced oscillations. **(Jun-2019) [NLJIET]**	**4**	**CO2**	**UN**
24.	Discuss in detail the forced vibration and amplitude resonance **(Jan-2024) [NLJIET]**	**4**	**CO2**	**UN**
25.	What is Damped Harmonic Motion? Derive differential equation for it. **(Aug-2022) [NLJIET]**	**4**	**CO2**	**AN**
26.	Define Reverberation time and write down the Sabine's formula of it by explaining the parameters in it. **(Jan-2020) [NLJIET]**	**4**	**CO2**	**UN**
27.	Write any four factors affecting acoustic of building and their remedies. **(July-2024) [NLJIET]**	**4**	**CO2**	**UN**
28.	Describe ultrasonic flaw detector. How is it used in detection of flows in metals? **(Mar-2021) [NLJIET]**	**4**	**CO2**	**RM**
29.	What is ultrasonic waves? Give properties and detection methods for Ultrasonics. **(Mar-2022) [NLJIET**	**4**	**CO2**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	What is Damped and Undamped vibrations? Derive the differential equation and general solution of damped harmonic motion. **(Jun-2019) [NLJIET]**	**7**	**CO2**	**AN**
2.	What is damping motion? Derive the differential equation and general solution of damped harmonic motion. **(Jan-2020) [NLJIET]**	**7**	**CO2**	**AN**
3.	What is damping motion? Derive the differential equation and general solution of damped harmonic motion. **(Jan-2024) [NLJIET]**	**7**	**CO2**	**AN**

4.	Describe Simple Harmonic motion with some examples. Differentiate Free and Forced oscillations (any four points). **(Mar-2021) [NLJIET]**	**7**	**CO2**	**RM**
5.	What is simple harmonic motion? Obtain equation for kinetic energy and potential energy of simple harmonic oscillator. Show that at any time the sum of kinetic energy and potential energy remains constant. **(Mar-2023) (Jan-2025-New) [NLJIET]**	**7**	**CO2**	**UN**
6.	Derive the differential equation of simple harmonic motion, also write its general solution and obtain the equation of velocity and acceleration of simple harmonic oscillator. **(Jun-2025-New) [NLJIET]**	**7**	**CO2**	**AN**
7.	Define piezoelectric effect and explain in detail piezoelectric ultrasonic generator with necessary circuit diagram. Also mention its merit and demerit. **(Jun-2019) [NLJIET]**	**7**	**CO2**	**UN**
8.	Explain in detail the production of ultrasonic waves through Piezoelectric oscillator method. **(Mar-2022) [NLJIET]**	**7**	**CO2**	**UN**
9.	(i) Write principle of Piezoelectric Generator. (ii) With proper circuit diagram describe construction and working of Piezoelectric Generator. (iii) List merits and demerits of Piezoelectric Generator. **(July-2024) [NLJIET]**	**7**	**CO2**	**RM**
10.	Describe production of ultrasonic waves by Piezo-electric method. Give its advantages and limitations. **(Jan-2025-New) [NLJIET]**	**7**	**CO2**	**RM**
11.	Explain the construction and working of piezoelectric method of ultrasonic wave production with suitable diagram. **(Jun-2025-New) [NLJIET]**	**7**	**CO2**	**UN**
12.	What are the applications of Ultrasound? Discuss them in detail. **(Mar-2022) [NLJIET]**	**7**	**CO2**	**UN**
13.	What is NDT? Write down 3 advantages and 3 disadvantages of NDT. **(Aug-2022) [NLJIET]**	**7**	**CO2**	**RM**

14.	Discuss Ultrasonic flaw detector method for NDT, with its advantages and limitation **(Aug-2023) [NLJIET]**	**7**	**CO2**	**UN**
15.	Explain in detail about the methods for the detection of ultrasound. Give any three applications of ultrasonic waves. **(Jan-2024) [NLJIET]**	**7**	**CO2**	**UN**
16.	Describe the different methods used for detection of ultrasonic waves. **(Mar-2023) [NLJIET]**	**7**	**CO2**	**RM**
17.	Write down the factors affecting the acoustics of an auditorium. Give remedies. **(Jan-2020) [NLJIET]**	**7**	**CO2**	**UN**
18.	State the acoustic requirements of a good auditorium. Write factors affecting architectural acoustics with remedies. **(Mar-2021) [NLJIET]**	**7**	**CO2**	**RM**
19.	What are the factors affecting acoustics of the building and give their remedies. **(Mar-2022) [NLJIET]**	**7**	**CO2**	**UN**
20.	Mention five factors affecting acoustic of building. Explain each and also mention their remedies. **(Aug-2022) [NLJIET]**	**7**	**CO2**	**UN**
	Examples (WAVES AND MOTION)			
1.	Find the acceleration of particle performing simple harmonic motion (SHM) when it is at 0.6 m from its mean position. The time period of SHM is 0.05 sec. Also calculate maximum velocity if the amplitude of SHM is 2 m. **(Jun-2019) [NLJIET]**	**3**	**CO2**	**AN**
2.	A particle execute simple harmonic motion with amplitude of 0.1 m and a period of 0.5 s. Calculate its displacement, velocity and acceleration (1/12) s after it crosses the mean position. **(Jan-2024) [NLJIET]**	**3**	**CO2**	**AP**
3.	An oscillator is performing SHM with an acceleration of 0.2 m/s^2 when displacement is 5 cm. Its maximum speed is 0.15 m/s. Find time period and amplitude of oscillator. **(Aug-2022) [NLJIET]**	**4**	**CO2**	**AN**

4.	Find the acceleration of particle performing simple harmonic motion when it is at 0.6 m from its mean position. The time period of S.H.M. is 0.05 sec. also calculate maximum velocity if the amplitude of S.H.M. is 2 m. **(Aug-2023) [NLJIET]**	**4**	**CO2**	**AN**
5.	A particle of mass 0.50 kg executes a simple harmonic motion. If it crosses the centre of oscillation with a speed of 10 m/s, find the amplitude of the motion. (force constant k = 50 N/m). **(Jun-2025-New) [NLJIET]**	**4**	**CO2**	**AP**
	Examples(ACOUSTICS AND ULTRASONICS)			
1.	An ultrasonic source of 150 KHz sends down a pulse towards the seabed, which returns after 0.82 s. The velocity of ultrasound in sea water is 1700 m/s. Calculate the depth of sea and wavelength of pulse. **(Jun-2019) [NLJIET]**	**3**	**CO2**	**AP**
2.	The volume of room is 800 m^3. The wall area of the room is 230 m^2, the floor and ceiling area is 130 m^2. The absorption coefficient of wall, floor and ceiling is 0.05, 0.75, and 0.09, respectively. Calculate the total absorption, average absorption coefficient and the reverberation time. **(Jun-2019) [NLJIET]**	**3**	**CO2**	**AP**
3.	Calculate length of an iron rod which can be used to produce ultrasound of frequency 22 kHz. Given that Young's Modulus and density of iron is 11.6×10^{10} N/m^2 and 7.25×10^3 kg/m^3, respectively. **(Jun-2019) [NLJIET]**	**3**	**CO2**	**AP**
4.	Calculate length of an iron rod which can be used to produce ultrasound of frequency 22 kHz. Given that Young's Modulus and density of iron is 11.6×10^{10} N/m^2 and 7.25×10^3 kg/m^3, respectively. **(Jun-2019) [NLJIET]**	**3**	**CO2**	**AP**
5.	An empty assembly hall have a volume of 11260 m^3 and its total absorption is equivalent to 92 m^2 of open window. What will be the effect on reverberation time if an audience fills the assembly hall full and thereby increase the absorption by another 92 m^2 of open window? **(March-2021) [NLJIET]**	**3**	**CO2**	**AP**

6.	Calculate the frequency to which piezoelectric oscillator circuit should be tuned so that a piezoelectric crystal of thickness 0.2 cm vibrates in it's fundamental mode to generate ultrasonic waves. Young's modulus is 80 Gpa and density of material is 2654 kg/m^3 **(March-2022) [NLJIET]**	**3**	**CO2**	**AN**
7.	A hall has a volume of 2790 m^3. It's total absorption is equivalent to 98.80 m^2 of open window. What will be the effect on reverberation time if the audience fill the hall and thereby increase the absorption by another 98.80 m^2 of open window? **(March-2022) [NLJIET]**	**3**	**CO2**	**AP**
8.	A hall, having volume of 6000 m^3, is found to have a reverberation time of 2.2 s. If the area of sound absorbing surface is 500 m^2, calculate the absorption coefficient. **(Aug-2022) [NLJIET]**	**3**	**CO2**	**AN**
9.	A piezoelectric X-cut crystal plate has a thickness of 1.6 mm. If the velocity of propagation of sound waves along the X-direction is 5760 m/s, calculate the fundamental frequency of the crystal. **(March-2023) [NLJIET]**	**3**	**CO2**	**AN**
10.	An ultrasonic source of 0.09 MHz sends down a Pulse towards the seabed which returns after 0.55 sec. The velocity of sound in water is 1800 m/s. Calculate the depth of the sea and wavelength of pulse. **(Aug-2023) [NLJIET]**	**3**	**CO2**	**AN**
11.	Find the frequency of the first and second modes of vibration for a quartz crystal of piezoelectric oscillator. The velocity of longitudinal waves in quartz is 5.5 x 10^3 m/s. thickness of quartz crystal is 0.05 m. **(Aug-2023) [NLJIET]**	**3**	**CO2**	**AN**
12.	A hall has a volume of 1,20,000 m^3, it has reverberation time of 1.5 sec. what is average absorbing power of the surface if the total absorbing. surface area is 25,000 m^2. **(Aug-2023) [NLJIET]**	**3**	**CO2**	**AP**

13.	An ultrasonic source of 0.09 MHz sends down a Pulse towards the seabed which returns after 0.55 sec. The velocity of sound in water is 1800 m/s. Calculate the depth of the sea and wavelength of pulse. **(Jan-2025-New) [NLJIET]**	**3**	**CO2**	**AP**
14.	(i) A cinema hall has a volume of 9,500 m^3. What should be the total absorption in the hall if the reverberation time of 1.7 s is to be maintained? (ii) An ultrasonic source of 0.075 MHz sends down a pulse towards the seabed, which returns after 0.95 s. The velocity of ultrasound in sea water is 1800 m/s. Calculate the depth of the sea and wavelength of pulse. **(March-2022) [NLJIET]**	**4**	**CO2**	**AP**
15.	The volume of a room is 1500 m^3. The wall area of the room is 240 m^2, the floor area is 130 m^2, and the ceiling area is 130 m^2. The average sound absorption coefficient (i) for wall is 0.035; (ii) for the ceiling is 0.75; and (iii) the floor is 0.05. Calculate the average sound absorption coefficient and reverberation time. **(Aug-2022) [NLJIET]**	**4**	**CO2**	**AN**
16.	An ultrasonic source of 0.09 MHz sends down a pulse towards the seabed which returns after 0.55s. The velocity of sound in water is 1800 m/s. Calculate (i) the depth of the sea and (ii) wavelength of pulse. **(Aug-2022) [NLJIET]**	**4**	**CO2**	**AN**
17.	A hall has a volume of 2265.6 m^3. Its total absorption is equivalent to 92.92 OWU. What will be the effect on the reverberation time if curtains of total absorption equivalent to 92.90 OWU are placed inside? **(March-2023) [NLJIET]**	**4**	**CO2**	**AP**
18.	Calculate the value of capacitance required to produce ultrasonic waves of 10^8 Hz with an inductance of 1 Henry. **(March-2023) [NLJIET]**	**4**	**CO2**	**AN**

19.	Calculate the length of iron rod which can be used to produce ultrasonic waves of 75 kHz. Given that the density of iron is 7230 kg /m^3 and its Young's modulus is 6×10^9 N/ m^2. **(July-2024) [NLJIET]**	**4**	**CO2**	**AN**
20.	The volume of a room is 1500 m^3.The wall area of the room is 240 m^2, the floor area is 125 m^2, and the ceiling area is 125 m^2. The average sound absorption coefficient (i) for wall is 0.025; (ii) for the ceiling is 0.85; and (iii) the floor is 0.09. Calculate the average sound absorption coefficient and reverberation time. **(July-2024) [NLJIET]**	**4**	**CO2**	**AN**
21.	Calculate the length of nickel rod required to produce ultrasonic waves of frequencies of 50 kHz and 100 kHz. Consider the fundamental mode of vibrations of the rod. Given, density of nickel= 8908 kg/m^3, Young's modulus of the nickel= 2.14×10^{11} N/m^2. **(Jan-2024) [NLJIET]**	**4**	**CO2**	**AP**
22.	The volume of a hall is 475 m^3. The area of wall is 200 m^2, area of floor and ceiling each is 100 m^2. If the absorption co-efficient of wall, ceiling and floor are 0.025, 0.02 and 0.55, respectively. Calculate the reverberation time for the hall. **(Jan-2025-New) [NLJIET]**	**4**	**CO2**	**AN**
23.	The volume of a room is 1500 m^3. The wall area of the room is 250 m^2, the floor and the ceiling area is 150 m^2 each. The average sound absorption coefficient for the wall, floor and the ceiling are 0.25, 0.06 and 0.80, respectively. Calculate the average absorption coefficient and the reverberation time. **(Jun-2025-New) [NLJIET]**	**4**	**CO2**	**AP**
24.	(i)Discuss the method to determine depth of the sea with the help of SONAR. (ii) An ultrasonic wave of 7.2 MHz sends down a pulse towards the seabed, which returns after 1.2 s. The velocity of sound in seawater is 1800 m/s. Calculate the depth of sea and wavelength of pulse. **(July-2024) [NLJIET]**	**7**	**CO2**	**AN**

CHAPTER: 3

OPTICS

- Huygens' Principle: Fundamental principle for wavepropagation.
- Superposition of Waves: Basic principle for understandinginterference and diffraction.
- Explanation of constructive and destructive interference.
- Applications in thin film interference, such as soap bubblesand oil films.
- Young's double slit experiment.
- Newton's rings,
- Michelson Interferometer
- Anti-reflection coating.
- Fresnel and Fraunhofer diffraction– diffraction due to 'n'slits- plane transmission grating.
- Rayleigh criterion for limit of resolution - resolving power ofgrating.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Explain Huygens' Principle for wavepropagation. **[NLJIET]**	3	CO3	UN
2.	Explain in detail soap bubbles and oil films. **[NLJIET]**	3	CO3	UN
3.	What is Anti-reflection coating? Explain in detail. **[NLJIET]**	3	CO3	UN
4.	Explain Huygens' Principle: Fundamental principle for wave propagation. **[NLJIET]**	3	CO3	UN
5.	State any three differences between interference and diffraction. **(Jan-2025-New) [NLJIET]**	3	CO3	RM
6.	State and explain principle of superposition of waves. **(Jan-2025-New) [NLJIET]**	3	CO3	RM
7.	Elaborate Explanation of constructive and destructive interference. **[NLJIET]**	4	CO3	AN
8.	What are the applications of Michelson's interferometer? **[NLJIET]**	4	CO3	AP
9.	Explain Rayleigh criterion for resolution. **[NLJIET]**	4	CO3	UN

10.	Distinguish between Fresnel and Fraunhofer diffraction. **[NLJIET]**	**4**	**CO3**	**UN**
11.	Differentiate Interference and Diffraction phenomena of light. **[NLJIET]**	**4**	**CO3**	**UN**
12.	Explain Superposition of Waves for understanding interference and diffraction. **[NLJIET]**	**4**	**CO3**	**UN**
13.	What is interference? Differentiate Constructive and destructive interference. **(Jan-2025-New) [NLJIET]**	**4**	**CO3**	**AN**
14.	Define constructive and destructive interference with their condition in terms of path difference.**(Jun-2025-New) [NLJIET]**	**4**	**CO3**	**RM**
15.	Define: (i) Interference of light, (ii) Diffraction of light, (iii) Fresnel diffraction. Also state Huygens' principle. **(Jun-2025-New) [NLJIET]**	**4**	**CO3**	**RM**

Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1.	Explain Young's double slit experiment in detail. **[NLJIET]**	**7**	**CO3**	**UN**
2.	Define Interference phenomena of light and Explain Young's double slit experiment of interference phenomenon of light using necessary diagram. **[NLJIET]**	**7**	**CO3**	**UN**
3.	Derive an expression for the radius of curvature in the Newton's ring experiment. **[NLJIET]**	**7**	**CO3**	**UN**
4.	Explain the formation of Newton's rings with suitable diagram and derive the equation of radius of bright and dark rings produced by the interference of reflected light. **(Jun-2025-New) [NLJIET]**	**7**	**CO3**	**AN**
5.	Explain construction and working of Michelson's Interferometer. **[NLJIET]**	**7**	**CO3**	**AN**
6.	Describe the principle, construction and working of Michelson interferometer. **(Jan-2025-New) [NLJIET]**	**7**	**CO3**	**AN**
7.	Define Diffraction. Explain the types of diffraction phenomena of light. **[NLJIET]**	**7**	**CO3**	**UN**

8.	Give construction and theory of a plane transmission diffraction grating. **[NLJIET]**	**7**	**CO3**	**UN**
9.	Obtain the formula for the resolving power of grating. **[NLJIET]**	**7**	**CO3**	**UN**
	Examples			
1.	Distance between two slits is 0.1 mm and the width of the fringes formed on the screen is 5 mm. If the distance between screen and slit is one meter calculate the wavelength of the light used. **[NLJIET]**	**3**	**CO3**	**AP**
2.	In Newton's ring experiment, the diameter of a certain bright ring is o.65 cm, and that of tenth ring beyond is 0.95 cm. If λ= 6000 A˚, calculate radius of curvature of a convex lens surface in contact with the glass plate. **[NLJIET]**	**3**	**CO3**	**AP**
3.	If the diameter of 5^{th} and 14^{th} dark rings are 0.525 cm and 0.900 cm respectively in a Newton's rings experiment, find the diameter of 23^{rd} ring. **[NLJIET]**	**3**	**CO3**	**AP**
4.	In Na yellow doublet has wavelengths 5890 A˚ and 5896 A˚. What should be the R.P. of a grating to resolve these lines? **[NLJIET]**	**3**	**CO3**	**AP**
5.	In a Young's double slit experiment, the two slits, which are 0.5 mm apart are illuminated by the yellow light of wavelength 589.0 nm. Calculate the distance between two consecutive bright bands in the interference pattern observed on a screen 2 m away from the slits. **(Jun-2025-New) [NLJIET]**	**3**	**CO3**	**AP**
6.	Newton's rings are observed in reflected light of λ = 5900 A˚. The diameter of the 5^{th} dark ring is 0.4 cm. Find the radius of curvature of the lens and the thickness of the air film. **[NLJIET]**	**4**	**CO3**	**AP**
7.	Calculate the minimum number of lines in a grating which will just resolve the lines of wavelength 5890 Å and 5896 Å in the second order. **(Jan-2025-New) [NLJIET]**	**4**	**CO3**	**AP**

CHAPTER: 4

QUANTUM PHYSICS

- Black body Radiation-Planck's law
- Energy distribution function,
- Wave-particle duality-de Broglie matter waves
- Concept of the wave function and its physical significance –Heisenberg's Uncertainty Principle
- Schrodinger's wave equation – Time-independent and Time-dependent equations
- Particle in a one-dimensional rigid box – tunneling(Qualitative) – Scanning Tunnelling Microscope.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	What is black body radiation? Explain. **[NLJIET]**	3	CO4	UN
2.	Write a short note on black body radiation. **(Jan-2025-New) [NLJIET]**	3	CO4	AN
3.	Write down the properties of Matter Waves. **[NLJIET]**	3	CO4	RM
4.	What is de-Broglie wavelength? How does it vary with momentum of the particle? **[NLJIET]**	3	CO4	UN
5.	What is de-Broglie wave? Write the equation of de-Broglie wavelength and mention the names of each quantity involved in the equation. **(Jun-2025-New) [NLJIET]**	3	CO4	RM
6.	What is Heisenberg's uncertainty principle? **[NLJIET]**	3	CO4	UN
7.	What do you understand by the wave function ψ of a moving particle? **[NLJIET]**	3	CO4	UN
8.	Discuss the energy distribution curve for a black body radiation with necessary diagram. **[NLJIET]**	4	CO4	UN
9.	What is photon? Mention the important properties of photon. **[NLJIET]**	4	CO4	RM
10.	Explain black body radiation for different temperature and wavelength with diagram. **[NLJIET]**	4	CO4	UN

11.	Define wave function. Write down the properties of photon. **[NLJIET]**	**4**	**CO4**	**RM**
12.	What is black body? Draw and explain nature of energy spectrum plot for black body radiation. **[NLJIET]**	**4**	**CO4**	**UN**
13.	With a schematic diagram, explain in brief the principle of scanning tunnelling microscope. **[NLJIET]**	**4**	**CO4**	**UN**
14.	Derive the expression for de Broglie wavelength for a particle when it is moving with Kinetic energy 'E'. **[NLJIET]**	**4**	**CO4**	**AN**
15.	For an electron accelerated by potential difference V, derive the expression for its dets deBroglie wave. **[NLJIET]**	**4**	**CO4**	**AN**
16.	Write a short note on Scanning Tunneling Microscope (STM). **(Jan-2025-New) [NLJIET]**	**4**	**CO4**	**AN**
17.	Write Schrodinger's time independent wave equation and explain its each term. Also write Heisenberg's uncertainty principle in terms of position and momentum. **(Jun-2025-New) [NLJIET]**	**4**	**CO4**	**UN**

DESCRIPTIVE QUESTIONS

	QUESTIONS	**Marks**	**CO**	**RBT Level**
1.	Derive the time dependent Schrodinger wave equation for particle having mass m and velocity v. **[NLJIET]**	**7**	**CO4**	**AN**
2.	Derive the time independent Schrodinger wave equation for particle having mass m and velocity v. **[NLJIET]**	**7**	**CO4**	**AN**
3.	Derive the time independent Schrödinger's wave equation. **(Jan-2025-New) [NLJIET]**	**7**	**CO4**	**AN**
4.	Discuss the Planck law for black body radiation with formula. **[NLJIET]**	**7**	**CO4**	**UN**
5.	Explain Heisenberg's uncertainty principle and derive it. **(Jan-2025-New) [NLJIET]**	**7**	**CO4**	**UN**

6.	Starting from the wave equation and introducing energy and momentum of the particle obtain an expression of three dimensional Schrodinger equation in time dependent form. **[NLJIET]**	**7**	**CO4**	**AN**
7.	Obtain three dimensional time dependent Schrodinger's wave equation from time independent Schrodinger's equation. **[NLJIET]**	**7**	**CO4**	**AN**
8.	Obtain Schrodinger's wave equation for a particle in square well potential and discuss energy levels when the well is infinitely deep. **[NLJIET]**	**7**	**CO4**	**UN**
9.	Explain tunnelling effect. Explain in brief how this is used in scanning tunnelling microscope. **[NLJIET]**	**7**	**CO4**	**UN**
10.	Explain the construction working of a scanning tunnelling microscope. **[NLJIET]**	**7**	**CO4**	**UN**
11.	Write a short-note on scanning tunneling microscope. **(Jun-2025-New) [NLJIET]**	**7**	**CO4**	**UN**
1.	Calculate the energy in eV for photon of wavelength 0.1×10^{-9} m. What is the momentum of this photon? **[NLJIET]**	**3**	**CO4**	**AN**
2.	Calculate the number of photons emitted by a 100 watt sodium lamp. The wavelength of emitted light is 5893 A˚.**[NLJIET]**	**3**	**CO4**	**AN**
3.	Find the energy of an electron moving in one dimension in an infinitely high potential box of width 0.1 nm.**[NLJIET]**	**3**	**CO4**	**AP**
4.	What is the lowest energy that a neutron can have if confined to move along the edge of an impenetrable box of length 10^{-14} m? (Given mass of neutron = 1.67×10^{-27} kg). **[NLJIET]**	**3**	**CO4**	**AN**
5.	Find the lowest energy of an electron confined in a cubical box of each side 1.5 A˚. **[NLJIET]**	**3**	**CO4**	**AP**
6.	Calculate the minimum energy an electron can possess in an infinitely deep potential well of width 4 nm. **[NLJIET]**	**3**	**CO4**	**AN**

7.	Calculate the lowest energy of the system consisting of three electron in a one dimension potential box of length 1 nm. **[NLJIET]**	**3**	**CO4**	**AN**
8.	Calculate the De-Broglie wavelength of a 20 keV neutron. Give mass of neutron, m= 1.67×10^{-27} kg. **[NLJIET]**	**3**	**CO4**	**AN**
9.	Calculate the wavelength of a photon and an electron both having an energy 2.0 eV. **[NLJIET]**	**3**	**CO4**	**AN**
10.	If a proton is moving with velocity 2% of the velocity of light, calculate the De-Broglie wavelength. **[NLJIET]**	**3**	**CO4**	**AN**
11.	An electron is trapped in an infinite potential wall of width 1.75 A˚. Calculate energy difference between ground and first energy level. **[NLJIET]**	**3**	**CO4**	**AN**
12.	Calculate the energy difference between the ground state and first excited state of an electron in the rigid box of length 1 A˚. **[NLJIET]**	**3**	**CO4**	**AN**
13.	Calculate de-Broglie wavelength associated with (i) A man of 50 kg mass walking with the speed of 2 m/s, and (ii) An electron moving with the speed of 3×10^{5} m/s. Based on your answers obtained in above two cases, explain why the wave nature of the man can't be observed. ($h = 6.63\times10^{-34}$ J·s, electron mass = 9.11×10^{-31} kg) **(Jun-2025-New) [NLJIET]**	**3**	**CO4**	**AP**

CHAPTER: 5

LASERS

- Properties of Laser, Einstein's theory of matter radiation: Aand B coefficients,
- Amplification of light by population inversion,
- Different types of lasers,
- Gas lasers (He-Ne),
- Solid-state lasers (Ruby laser),
- Properties of laser beams: Mono-chromaticity, coherence,directionality and brightness, laser speckles,
- Applications of lasers in science, engineering, and medicine.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Write full for of LASER and discuss in brief characteristics of laser light. **(July-2024) [NLJIET]**	3	CO5	UN
2.	Write down the properties of LASER light. **(Jan-2020) [NLJIET]**	3	CO5	RM
3.	Explain various properties of LASER beam. **(Mar-2022) [NLJIET]**	3	CO5	UN
4.	Write and explain any three properties of LASER. **(Aug-2022) [NLJIET]**	3	CO5	UN
5.	Discuss any three important applications of LASER. **(Jan-2024) [NLJIET]**	3	CO5	UN
6.	Write any three applications of LASER. **(July-2024) [NLJIET]**	3	CO5	AP
7.	Describe and differentiate ordinary light and Laser light. **(Mar-2021) [NLJIET]**	3	CO5	RM
8.	Describe population inversion with suitable diagram. **(Mar-2021) [NLJIET]**	3	CO5	RM

9.	Describe role of stimulated emission in lasing action. **(Mar-2021) [NLJIET]**	**3**	**CO5**	**RM**
10.	Write the advantages of using laser drilling in industries. **(Mar-2023) [NLJIET]**	**3**	**CO5**	**UN**
11.	Define Spontaneous and Stimulated emission. **(Aug-2023) [NLJIET]**	**3**	**CO5**	**RM**
12.	Define:(1) Meta stable state (2) Population inversion (3) Pumping **(Jan-2025-New) [NLJIET]**	**3**	**CO5**	**RM**
13.	What is the full form of LASER? Give applications of Laser in various fields. **(Jun-2019) [NLJIET]**	**4**	**CO5**	**AP**
14.	Derive the relation between Einstein's coefficients with necessary assumptions. **(Jun-2019) [NLJIET]**	**4**	**CO5**	**AN**
15.	Derive the relationship between Einstein Coefficients. **(Jan-2020) [NLJIET]**	**4**	**CO5**	**AN**
16.	Define Optical resonator, life time, metastable state and Pumping mechanism for LASER. **(Mar-2022) [NLJIET]**	**4**	**CO5**	**RM**
17.	Write properties of Laser light. **(Mar-2021) [NLJIET]**	**4**	**CO5**	**UN**
18.	Write the properties of laser and explain in brief. **(Jun-2025-New) [NLJIET]**	**4**	**CO5**	**UN**
19.	Compare spontaneous emission and stimulated emission. (Mention four points of comparison) **(Aug-2022) [NLJIET]**	**4**	**CO5**	**AN**
20.	What is population inversion? Explain the different methods for achieving population. **(Mar-2023) [NLJIET]**	**4**	**CO5**	**UN**
21.	What is population inversion (draw suitable diagram)? Explain why it is necessary for the production of laser? **(Jun-2025-New) [NLJIET]**	**4**	**CO5**	**UN**
22.	Explain the term (i) stimulated emission, (ii) population inversion, (iii) pumping and (iv) metastable state. **(Jan-2024) [NLJIET]**	**4**	**CO5**	**UN**

23.	Differentiate: Spontaneous and Stimulated emission. **(Jan-2025-New) [NLJIET]**	**4**	**CO5**	**AN**
DESCRIPTIVE QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1.	Write the full form of laser and explain (i) Absorption, (ii) Spontaneous emission, and (iii) Stimulated emission with suitable diagrams. **(Jun-2025-New) [NLJIET]**	**7**	**CO5**	**UN**
2.	Explain in detail construction and working of Ruby Laser with the help of necessary schematic and energy level diagram. **(Jun-2019) [NLJIET]**	**7**	**CO5**	**UN**
3.	Write down the various applications of LASER. **(Jan-2020) [NLJIET]**	**7**	**CO5**	**AP**
4.	Give applications of LASER in various fields in detail. **(Mar-2022) [NLJIET]**	**7**	**CO5**	**AP**
5.	Write applications of LASER in various field. **(Aug-2022) [NLJIET]**	**7**	**CO5**	**AP**
6.	What is the full form of LASER? Explain in detail construction and working of Ruby Laser with the help of necessary schematic and energy level diagram. **(Mar-2021) [NLJIET]**	**7**	**CO5**	**UN**
7.	Explain construction and working of Ruby LASER. **(Aug-2022) [NLJIET]**	**7**	**CO5**	**UN**
8.	Explain with neat sketch the construction and working of ruby laser. **(Mar-2023) [NLJIET]**	**7**	**CO5**	**UN**
9.	Explain construction and working of Ruby laser with necessary diagrams. **(Aug-2023) [NLJIET]**	**7**	**CO5**	**UN**
10.	Explain in detail construction and working of Ruby LASER with necessary schematic and energy level diagrams. **(Jan-2024) [NLJIET]**	**7**	**CO5**	**UN**
11.	Write detailed note on Ruby laser with proper diagram. **(July-2024) [NLJIET]**	**7**	**CO5**	**UN**

12.	Establish the relation between Einstein's coefficients. **(Mar-2022) [NLJIET]**	**7**	**CO5**	**AP**
13.	What are Einstein's coefficients? Give relation between them and discuss the result. **(Jan-2025-New) [NLJIET]**	**7**	**CO5**	**AN**
14.	Obtain the relation between Einstein's coefficients. **(Jun-2025-New) [NLJIET]**	**7**	**CO5**	**AN**
15.	Explain in detail construction and working of He-Ne Gas LASER with necessary schematic and energy level diagrams. **(Mar-2022) [NLJIET]**	**7**	**CO5**	**UN**
16.	What is a gas laser? Explain the working of He-Ne laser with relevant diagrams. **(Mar-2023) [NLJIET]**	**7**	**CO5**	**UN**
17.	Explain construction and working of He – Ne laser. **(Aug-2023) [NLJIET]**	**7**	**CO5**	**UN**
18.	Explain basic components of laser generation with diagram. Also give types of laser. **(Aug-2023) [NLJIET]**	**7**	**CO5**	**UN**
19.	Write detailed note on He-Ne laser with proper diagram. **(July-2024) [NLJIET]**	**7**	**CO5**	**UN**
20.	What is full form of LASER? Write the characteristics and applications of LASER. **(Jan-2025-New) [NLJIET]**	**7**	**CO5**	**AN**
EXAMPLES				
1.	Calculate the wavelength of Laser light if the separation between metastable state and lower energy state is 1.80 eV. (Consider Planck's constant is 4.14×10^{-15} eV.s) **(Jun-2019) [NLJIET]**	**3**	**CO5**	**AN**
2.	What is the wavelength of light of Ruby Laser if the separation between metastable state and lower energy state is 1.79 eV. Given that Planck's constant = 6.64×10^{-34} Js. **(Jan-2020) [NLJIET]**	**3**	**CO5**	**AP**
3.	In Ruby laser, total number of Cr^{+3} ions in excited state are 2.8 $\times10^{19}$. If the laser emits radiation of wavelength 7000 A°, calculate energy of laser pulse. **(Mar-2023) [NLJIET]**	**3**	**CO5**	**AN**

4.	In He-Ne laser system, the two energy levels of Ne involved in lasing action have energy values of 20.66 eV and 18.70 eV. Population inversion occurs between these two levels. What will be the wavelength of a laser beam produced? What will be the population of the metastable energy level with respect to the upper excited level at 27 ºC? **(Jan-2024) [NLJIET]**	**3**	**CO5**	**AP**
5.	In He-Ne gas laser, Ne atoms emit laser photons of wavelength 632.8 nm while electrons make transitions from E_3 state to E_2 state. Find the energy difference between these two states. ($h = 6.63\times10^{-34}$ J·s and $c = 3\times10^{8}$ m/s) **(Jun-2025-New) [NLJIET]**	**3**	**CO5**	**AP**
6.	A ruby laser emits light of wavelength 694.3 nm. Calculate the frequency and the energy of the laser photons emitted. ($h = 6.63\times10^{-34}$ J·s and $c = 3\times10^{8}$ m/s) **(Jun-2025-New) [NLJIET]**	**3**	**CO5**	**AP**
7.	Using Einstein's theory, show that in the optical region, say, at λ=5000 A˚ and T=300 K, the amplification is not possible. **(Mar-2023) [NLJIET]**	**4**	**CO5**	**AP**

CHAPTER: 6

NEW ENGINEERING MATERIALS

SEMICONDUCTOR MATERIALS:

- Introduction of Group IV elements,
- Properties,
- Concept of carriers,
- Concept of bands and band gap,
- P-type, N-type materials
- Introduction to P-N Junction Diode and I-V characteristics,
- Zener diode and its characteristics

SUPERCONDUCTING MATERIALS:

- Introduction – Properties
- Meissner effect
- Type I & Type II superconductors
- Applications.

NANOMATERIALS:

- Introduction
- Synthesis of nanomaterials – Top down and Bottom upapproach
- Ball milling
- PVD method
- Applications.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Differentiate between intrinsic and extrinsic semiconductors. **(Jan-2019) [NLJIET]**	3	CO6	UN
2.	Differentiate between p-type and n-type semiconductors **(Jan-2019) [NLJIET]**	3	CO6	UN
3.	Explain n-type and p-type semiconductors with suitable diagrams. **(Mar-2021) [NLJIET]**	3	CO6	UN
4.	Write down any three differences between intrinsic and extrinsic semiconductors. **(Mar-2021) [NLJIET]**	3	CO6	UN
5.	Compare Intrinsic and Extrinsic Semiconductors. **(Mar-2023)[NLJIET]**	3	CO6	UN
6.	Explain /Discuss Meissner effect in superconductors. **(Jan-2019) (Jun-2019)[NLJIET]**	3	CO6	UN
7.	Explain the Josephson's junction and state any one of its application. **(Jan-2024)[NLJIET]**	3	CO6	UN
8.	Explain: SQUIDS and its applications. **(Mar-2021) [NLJIET]**	3	CO6	UN

9.	What is Meissner effect? For superconductor show that $\chi_m = -1$.) **(Jul-2024) [NLJIET]**	3	CO6	RM
10.	Define following terms. Magnetic Levitation, Persistent Current, Isotope effect **(Mar-2023) [NLJIET]**	3	CO6	RM
11.	Define: (1) Critical Temperature (2) Critical Magnetic field (3) Critical Current density. **(Aug-2022) (Jan-2025-New) [NLJIET]**	3	CO6	RM
12.	Define: Superconductivity and Transition temperature Tc. **(Aug-2023) [NLJIET]**	3	CO6	RM
13.	Show that superconductors are perfect diamagnet. **(Mar-2023) [NLJIET]**	3	CO6	AP
14.	Compare /Differentiate between Hard and Soft superconductors. **(Jun-2019) [NLJIET]**	3	CO6	UN
15.	Write any three properties of superconductors. **(May-2018) [NLJIET]**	3	CO6	RM
16.	Explain: SQUIDS and its applications. **(Mar-2021) [NLJIET]**	3	CO6	UN
17.	Give any six applications of nano-materials. **(Jan-2024) [NLJIET]**	3	CO6	RM
18.	Define nanomaterials and explain ball milling technique of synthesis of nanomaterials with suitable diagram. **(Jun-2025-New)[NLJIET]**	3	CO6	UN
19.	Identify three important factors that defines a superconducting state in detail. **(Mar-2023) [NLJIET]**	4	CO6	RM
20.	Differentiate: Intrinsic and Extrinsic Semiconductor. **(Aug-2023) [NLJIET]**	4	CO6	UN
21.	Explain intrinsic and extrinsic semiconductors with necessary diagrams. **(Jun-2019) [NLJIET]**	4	CO6	UN
22.	What is Zener diode? Explain with circuit diagram how a Zener diode operates in reverse bias condition. **(Jun-2019) [NLJIET]**	4	CO6	UN

23.	Write a short note on P-N junction diode. **(Mar-2022) [NLJIET]**	4	CO6	RM
24.	Write any four differences between n-type and p-type semiconductors. **(Jun-2025-New) [NLJIET]**	4	CO6	UN
25.	Compare Type-I and Type-II superconductor. **(Jan-2019) (Aug-2022) (Mar-2023) [NLJIET]**	4	CO6	UN
26.	Write any four points of comparison/ distinguish between Type–I and Type-II superconductors. **(Jun-2019)[NLJIET]**	4	CO6	UN
27.	Differentiate: Type I and Type II superconductors. **(Aug-2023) [NLJIET]**	4	CO6	UN
28.	What is SQUID? Explain with diagram? Give formation and application of SQUID. **(Jun-2019) (Jan-2020) [NLJIET]**	4	CO6	RM
29.	Write down applications associated with SQUID. **(Aug-2023) [NLJIET]**	4	CO6	RM
30.	Write down the applications of superconductors. **(Mar-2021) [NLJIET]**	4	CO6	RM
31.	Explain Josephson Junction and its application. Also explain application of superconductor in Cryotron. **(Jan-2020) [NLJIET]**	4	CO6	UN
32.	What is Josephson junction? Write a short note on SQUID. **(Mar-2022) [NLJIET].**	4	CO6	RM
33.	Write any two applications of nanomaterials and Biomaterials. **(Mar-2022) [NLJIET]**	4	CO6	RM
34.	Write disadvantages of Nanomaterials. **(Jun-2019) (Jan-2019) [NLJIET]**	4	CO6	RM
35.	List out the most important properties of nanomaterials. **[NLJIET]**	4	CO6	RM
36.	Write applications of nano materials. **(Jan-2025-New) [NLJIET]**	4	CO6	AP

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Explain intrinsic and extrinsic (N & P type) semiconductors with the help of energy band diagram. **(Mar-2022) [NLJIET]**	**7**	**CO6**	**UN**
2.	Explain how the materials are classified into conductors, semiconductors and insulators with the help of energy band diagrams. **(Mar-2023) [NLJIET**]	**7**	**CO6**	**UN**
3.	What is a PN Junction diode? What is external bias? Describe its forward and reverse bias conditions with appropriate diagrams. **(Jun-2019) [NLJIET]**	**7**	**CO6**	**RM**
4.	Explain: Working of the p-n junction diode and state its applications. **(Aug-2023) [NLJIET]**	**7**	**CO6**	**UN**
5.	Explain forward and reverse biasing of p-n junction diode. **(Jan-2020) [NLJIET]**	**7**	**CO6**	**UN**
6.	Explain formation and working of p-n junction diode in forward and reverse biasing by proper diagrams and its I-V characteristics.**(Mar-2021) [NLJIET]**	**7**	**CO6**	**UN**
7.	Explain working of the p-n junction diode and state its applications. **(Jan-2020) (Jan-2025-New)[NLJIET]**	**7**	**CO6**	**AN**
8.	Explain the Meissner effect and prove that superconductors are perfect diamagnetic. **(Aug-2023) [NLJIET]**	**7**	**CO6**	**UN**
9.	Discuss the properties of superconductors. **(Jan-2019) [NLJIET]**	**7**	**CO6**	**UN**
10.	Write down various properties of superconductors. **(Mar-2021)[NLJIET]**	**7**	**CO6**	**RM**
11.	Explain Meissener's effect in detail and show that for superconductor, $\chi_m = -1$. **(Mar-2022)[NLJIET]**	**7**	**CO6**	**UN**
12.	What is superconductivity? Explain any six properties of superconductor. **(Mar-2022)(Jul-2024)[NLJIET].**	**7**	**CO6**	**UN**

13.	Write and explain characteristics of superconductors. **(Aug-2022)[NLJIET]**	**7**	**CO6**	**UN**
14.	List out the techniques used in synthesis of nanomaterials. Discuss any two of them in detail. **(Mar-2023) (Mar-2022) (Jan-2020) [NLJIET]**	**7**	**CO6**	**UN**
15.	Explain Ball Milling technique for the production of nano particles with its merits and demerits. **(Jan-2019)[NLJIET]**	**7**	**CO6**	**UN**
16.	Explain Physical Vapour Deposition (PVD) method for synthesis of nanoparticles. **[NLJIET]**	**7**	**CO6**	**UN**
EXAMPLES				
	QUESTIONS	**Marks**	**CO**	**RBT Level**
1.	The critical temperature of Nb is 9.15 K. At zero Kelvin the critical field is 0.196 Tesla. Calculate the critical field at 6K. **(Jan-2019) [NLJIET]**	**3**	**CO6**	**AP**
2.	The critical current density equal to 1.71 x 10^8 A/m^2 is required to change a superconducting wire of radius 0.5 mm at 4.2 K. If the critical temperature of the material is 7.18 K, calculate the maximum value of the critical magnetic field. **(Jun-2019)[NLJIET]**	**3**	**CO6**	**AP**
3.	Calculate the critical current for a superconducting wire of lead having a diameter of 2 mm at 2K. Critical temperature for lead is 4 K and $H_c(0)$ = 6.5 X 10^4 A/m. **(Jan-2019- NEW)[NLJIET]**	**3**	**CO6**	**AP**
4.	At 0 magnetic field, a superconducting Tin has a critical temperature of 3.7 K. At 0 K, critical magnetic field is 0.306 T. Calculate the critical magnetic field at 2 K. **(Mar-2021)[NLJIET]**	**3**	**CO6**	**AP**
5.	Calculate the critical current for a superconducting wire of lead having a diameter of 1mm at 4.2 K. Critical temperature for lead is 7.18 K and $H_c(0)$ = 6.5 x 10^4 A/m. **(Mar-2022)[NLJIET]**	**3**	**CO6**	**AP**

6.	The critical temperature for Hg with isotopic mass 199.5 is 4.185 K. Calculate its critical temperature when its isotopic mass changes to 203.4.**(Aug-2022)[NLJIET]**	**3**	**CO6**	**AP**
7.	The critical current passing through a superconducting wire of radius 5 mm is 21 A. Calculate critical magnetic field. **(Jul-2024) [NLJIET]**	**3**	**CO6**	**AP**
8.	Transition temperature of mercury of mass number 202 is 4.20 K. Find the transition temperature of isotope of mercury of mass number 204. **(Jun-2025-New) [NLJIET]**	**3**	**CO6**	**AP**
9.	For specimen of V_3Ga, the critical fields are 1.4 x 10^5 A/m and 4.2 x 10^5 A/m at 14 K and 13 K respectively. Calculate the value of transition temperature. **(Aug-2022) [NLJIET]**	**4**	**CO6**	**AP**
10.	Calculate the critical current which can flow through a long thin superconducting wire of diameter 10^{-3} m. Given Hc=7.9 X 10^3 amp/m**. (Aug-2023)[NLJIET]**	**4**	**CO6**	**AP**
11.	Determine the critical current for a superconducting ring of diameter 10^{-3} m at temperature 4.2 K. Given the critical temperature (Tc) for sample is 7.18 K and critical magnetic field at 0 K (Hc(0)) is 6.5×10^4 A/m.. **(Jan-2024) [NLJIET]**	**4**	**CO6**	**AP**
12.	A critical magnetic field of lead wire is 6.5×10^3 A/m at 0 K. At what temperature would the critical magnetic field of lead drop to 4.5×10^3 A/m if the critical temperature of lead is 7.18 K? What is the critical current density at that temperature if the diameter of the wire is 2×10^{-3} m? **(Jan-2024) [NLJIET]**	**4**	**CO6**	**AP**
13.	The critical temperature for a metal with isotopic mass 199.5 is 4.185 K. Calculate the isotopic mass if the critical temperature falls to 4.133 K. **(Jan-2025-New) [NLJIET]**	**4**	**CO6**	**AP**

14.	(i) The critical field for a superconducting sample is 10^6 Am^{-1} at 9.58 K and 2 X 10^6 A m^{-1} at 0 K. Determine the Tc value. (ii) Calculate the critical current through a long thin superconducting wire of diameter 0.5 mm. The critical field is 9.2 kA/m. **(Mar-2023) [NLJIET]**	**7**	**CO6**	**AP**

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